

Security Assessment Report

Automated Modular Bash Scanning & Evidence Gathering Framework

TARGET SYSTEM	ASSESSMENT DATE	FRAMEWORK TYPE	ENVIRONMENT
DC-6 Machine (192.168.56.101)	September 26, 2026	Modular Bash Automation	Kali Linux (VirtualBox)

1. Executive Summary & Architecture

This document presents the architecture and execution results of an automated security scanning framework implemented in Bash. The solution performs automated host reachability checks, port discovery via Nmap, clean service parsing, and automated protocol-specific enumeration across HTTP, SSH, FTP, SMB, DNS, and SMTP modules.

Project Root Directory Structure

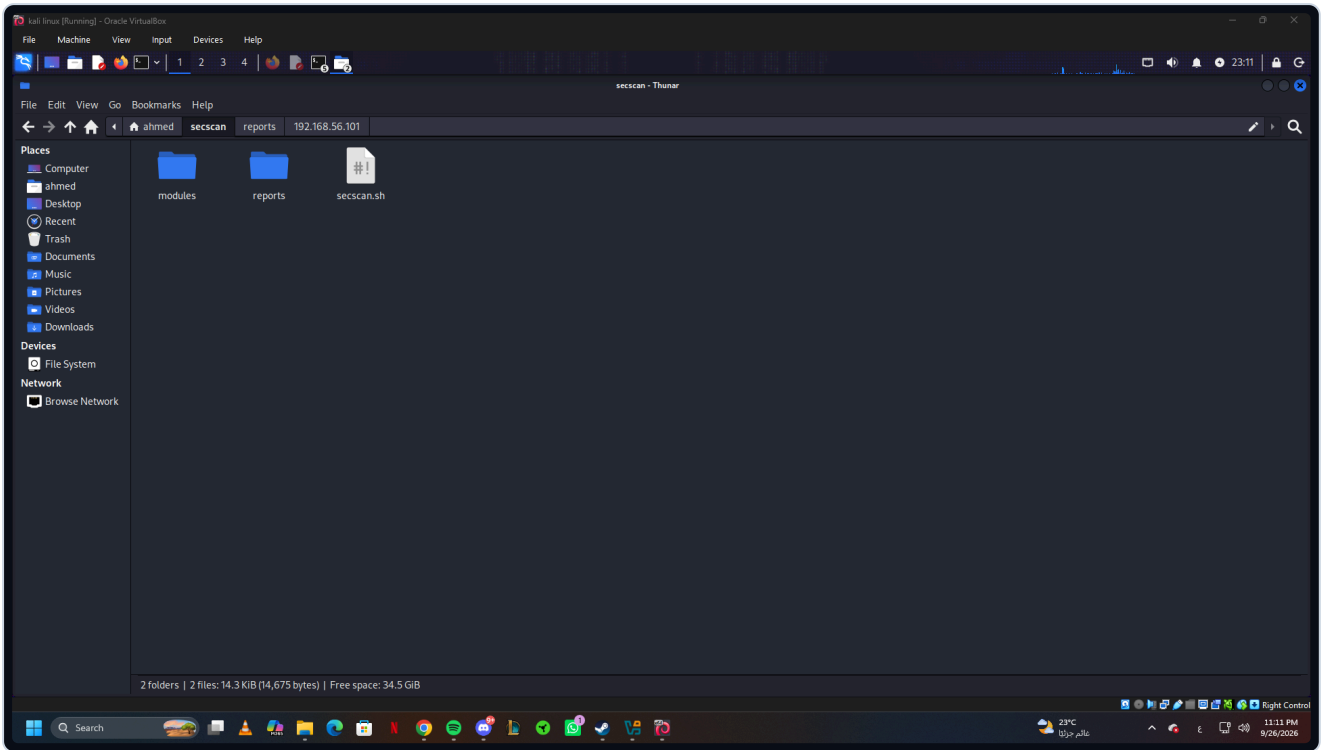
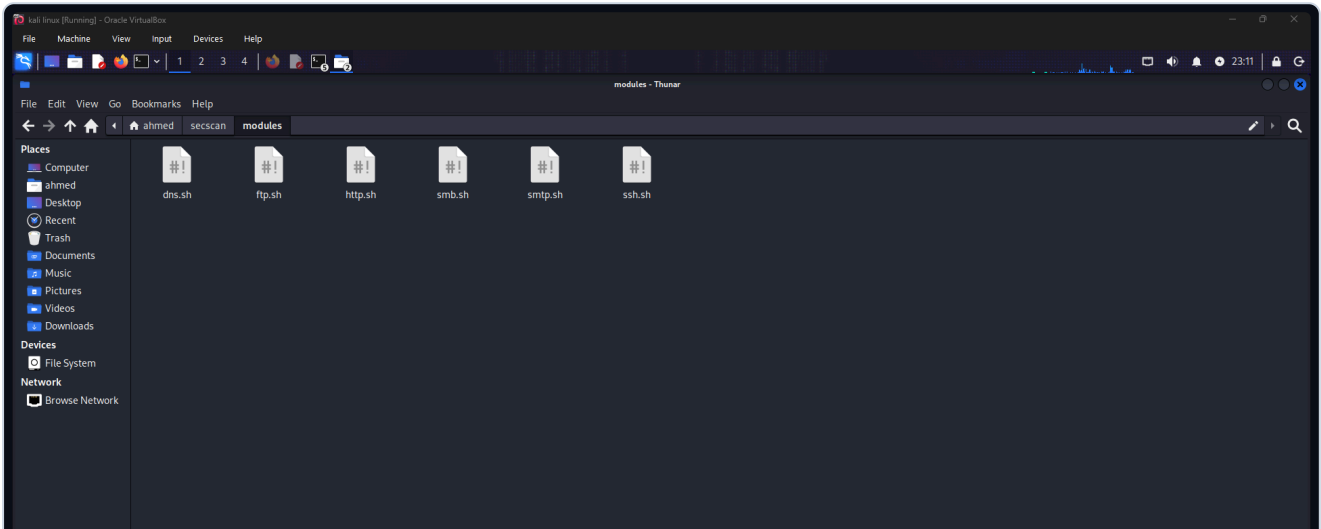


Figure 1.1: Root directory hierarchy displaying core controller script (secscan.sh), modules, and reports directory.

Service Modules Structure



2. Execution Workflow & Output Structure

The primary execution controller `secscan.sh` validates target availability before executing core service identification. Port parsing triggers corresponding dynamic sub-scripts to gather detailed evidence cleanly without manual intervention.

Core Controller Script Execution

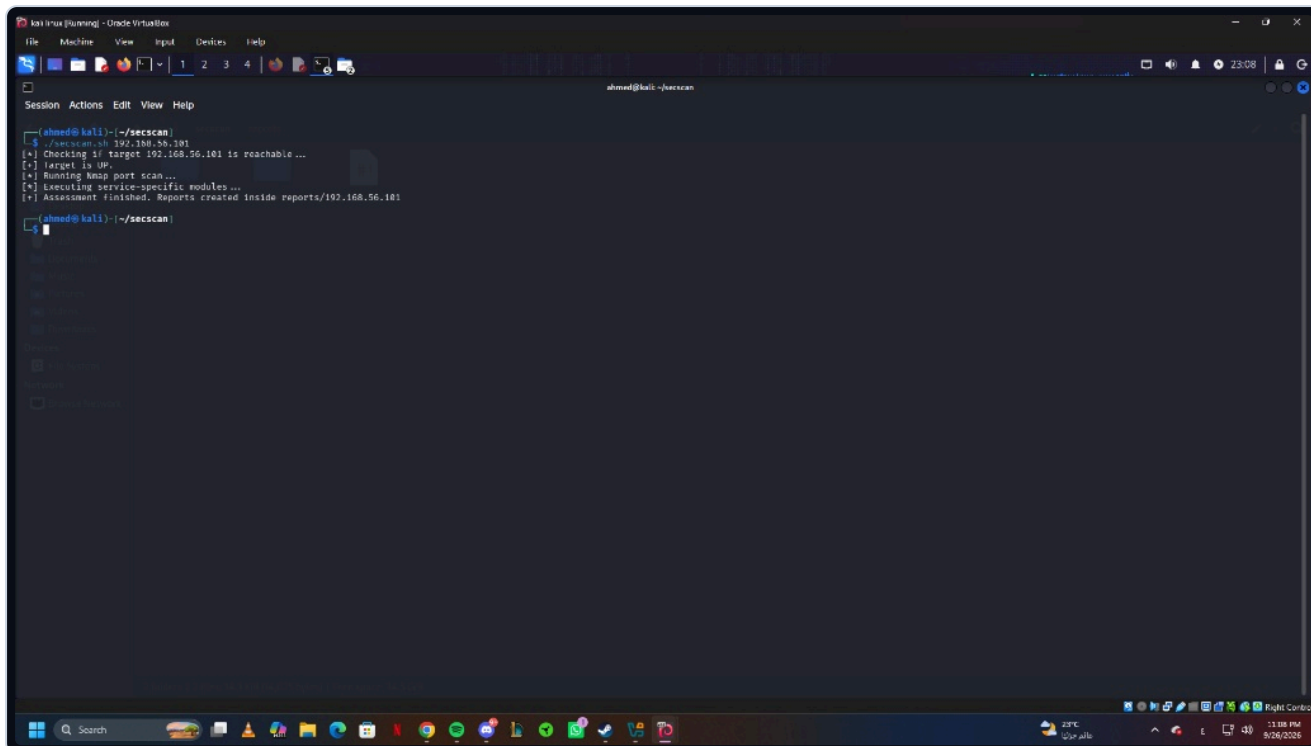


Figure 2.1: Terminal execution showing target connectivity check and assessment completion for 192.168.56.101.

Target Generated Report Directory

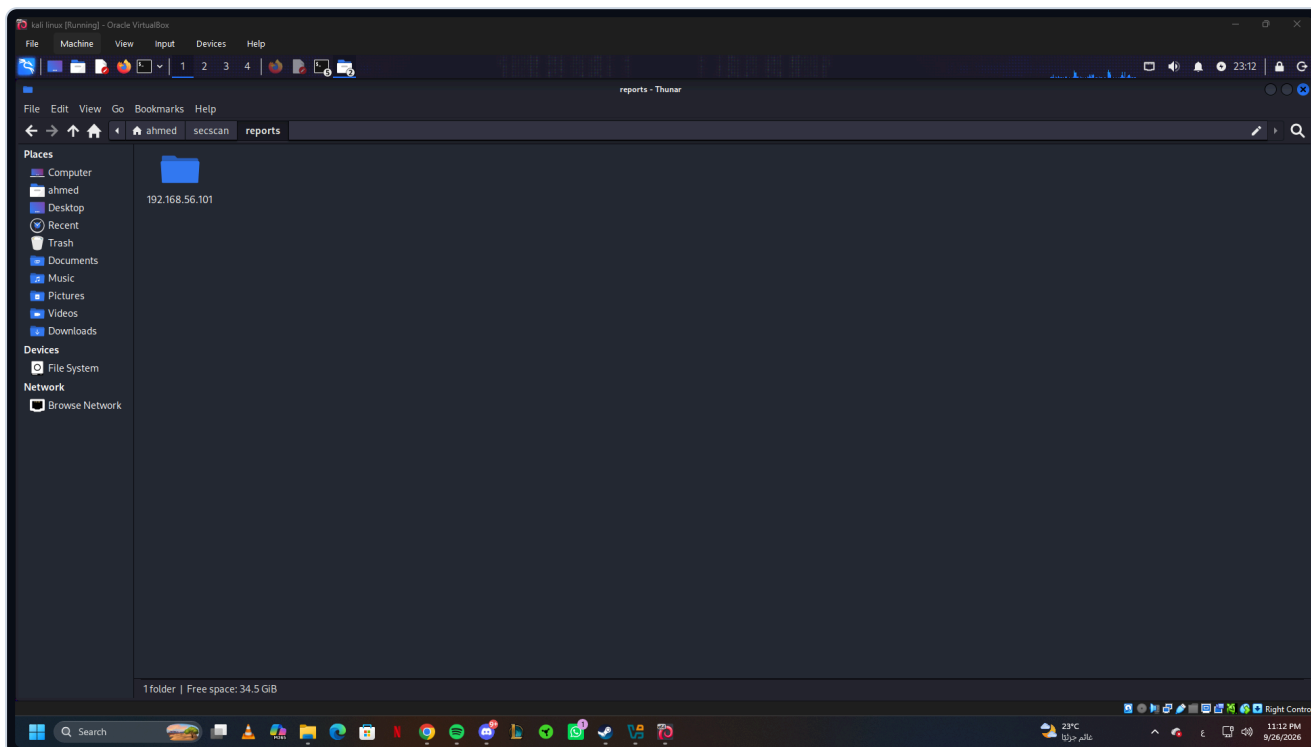


Figure 2.2: Automated creation of target-specific folder inside reports/192.168.56.101/.

Generated Evidence Artifacts



3. Security Findings & Technical Evidence

Port	State	Service	Detected Version	Module Automated Action
22/tcp	Open	SSH	OpenSSH 7.4p1 Debian 10	Retrieved SSH Banner via socket connection.
80/tcp	Open	HTTP	Apache httpd 2.4.25 (Debian)	Extracted HTTP response headers and checked robots.txt.

Summary Log Output (`summary.txt`)

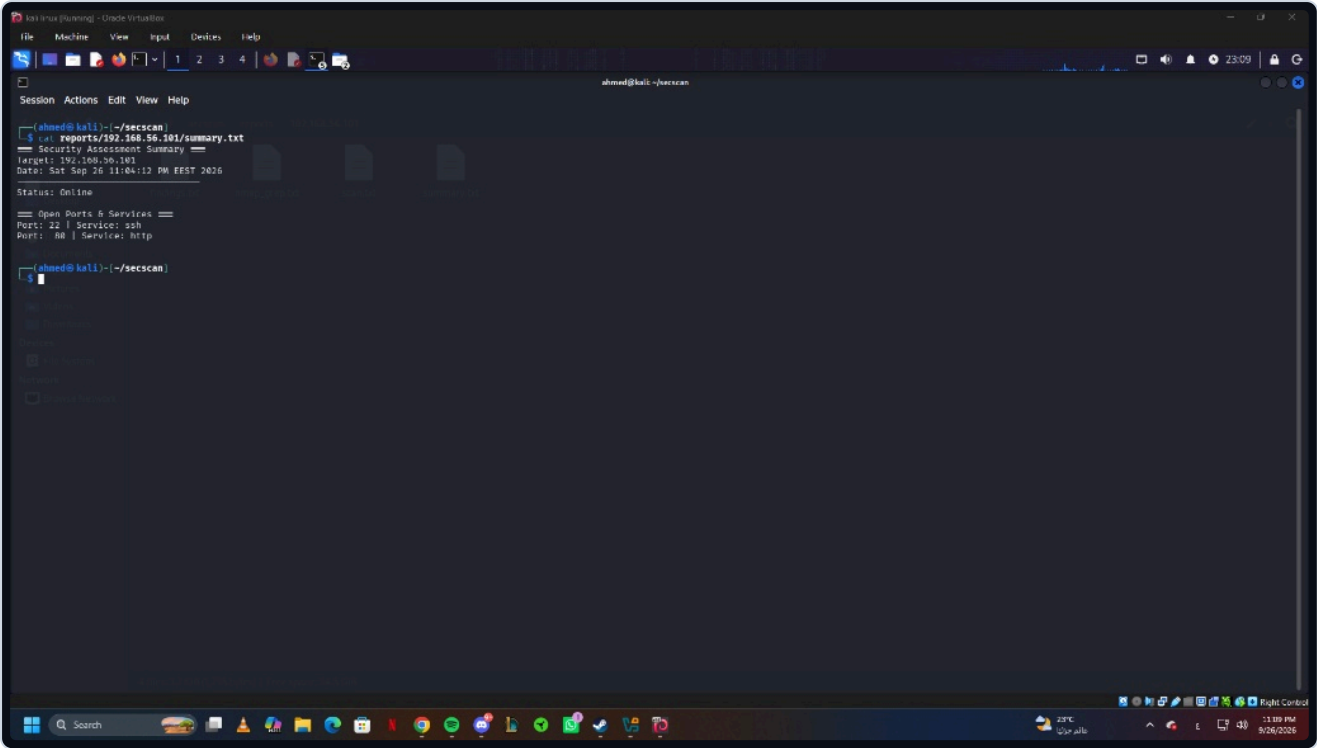


Figure 3.1: Clean structured summary detailing target status and identified open services.

Evidence & Findings Output (`findings.txt`)

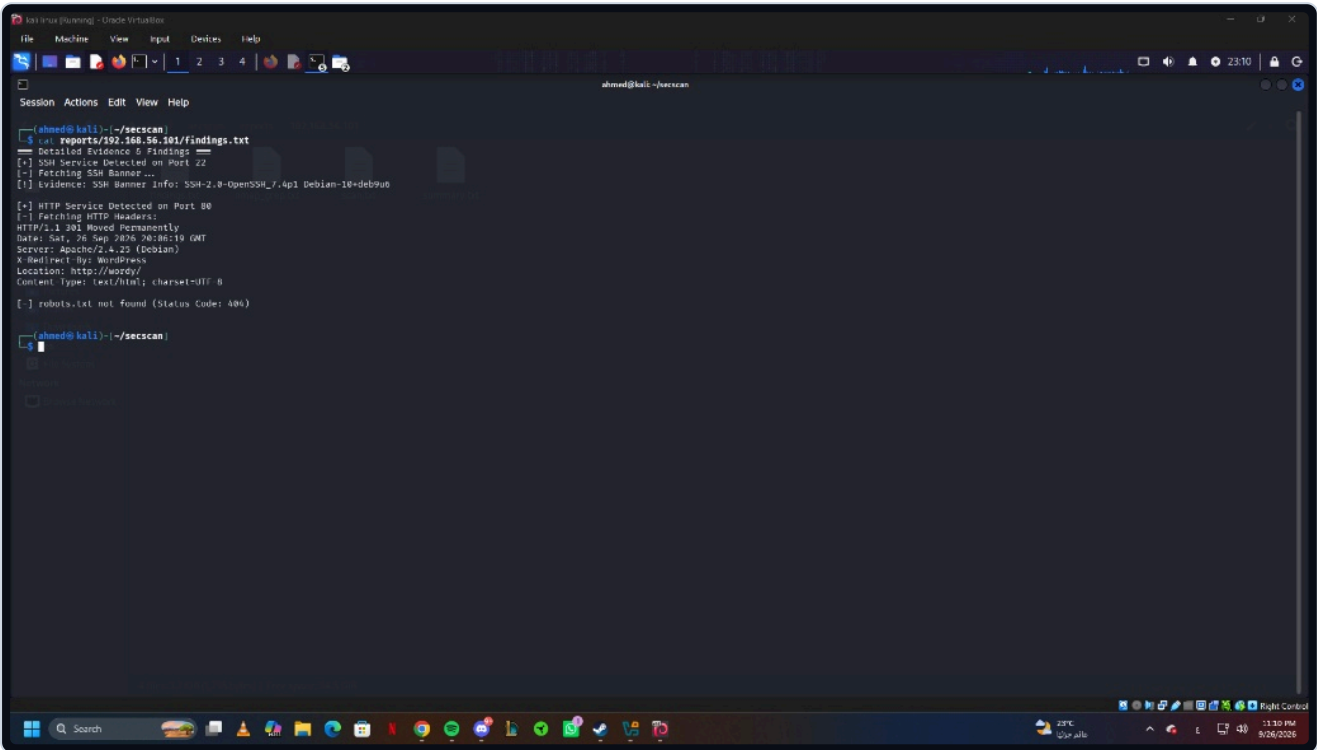


Figure 3.2: Detailed evidence output demonstrating automated extraction of SSH banner and HTTP headers.